Meta-analysis of deep-sequenced fungal communities indicates limited taxon sharing between studies and the presence of biogeographic patterns

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Summary

- High-throughput amplicon sequencing gives new insights into fungal community ecology. Massively generated molecular data lead to the discovery of vast fungal diversity. However, it is unclear to what extent operational taxonomic units (OTUs) overlap among independent studies, because no comparative studies exist.
- We compared fungal diversity based on the internal transcribed spacer (ITS1) region among 10 published studies. Starting from the raw 454 pyrosequencing data, we used a uniform pipeline to prune the reads. We investigated fungal richness and taxonomic composition among phyllosphere and soil fungal communities, as well as biogeographic signals in the data.
- We did not find globally distributed OTUs, even when comparing fungal communities from similar habitats (phyllosphere or soil). This suggests that high local fungal diversity scales up to high global diversity. The most OTU-rich classes in the phyllosphere were Dothideomycetes (21%) and Sordariomycetes (14%), and in the soil were Sordariomycetes (13%) and Agaricomycetes (12%). The richness estimates suggest the presence of undiscovered fungal diversity even in deeply sequenced study systems.
- The small number of OTUs shared among studies indicates that globally distributed taxa and habitat generalists may be rare. Latitudinal diversity decline and distance decay relationships suggest the presence of biogeographic patterns similar to those in plants and animals.

Introduction

High-throughput amplicon sequencing, or environmental metabarcoding, is rapidly gaining importance in fungal biodiversity research. It allows the generation of barcode markers of often unculturable fungal lineages at unprecedented magnitude. These data promise to revolutionize our understanding of global fungal diversity and function (Hibbett & Taylor, 2013). Insights into fungal community ecology that could be gained or confirmed with high-throughput sequencing methods include, for example, the effect of land use (Lumini *et al.*, 2010), host plant ecological group (Öpik *et al.*, 2009) and host plant genotype (Bálint *et al.*, 2013) on fungal community composition, the temporal and spatial structuring of mycorrhizal communities (Jumpponen *et al.*, 2010; Blaalid *et al.*, 2012), an unexpectedly high local diversity (Buée *et al.*, 2009), and the discovery of previously unknown fungal lineages (Amend *et al.*, 2012). However, comparative analyses of independent and spatially distant original studies of deep-sequenced fungal communities are still missing. Therefore, it is unclear whether operational taxonomic units (OTUs) overlap among different habitats, or whether large-scale biogeographic signals are present in high-throughput metabarcoding data. It is important to robustly assess fungal biogeographic patterns, including the identification of generalist taxa across habitats and geographic regions, or the presence of latitudinal gradients, given the strong interactive effects between microbial and plant communities (Friesen *et al.*, 2011; Peay *et al.*, 2013).

One reason for the lack of comparative high-throughput metabarcoding studies is that OTUs reported in independent original studies are not directly comparable. Different researchers use different methods for data generation, data pruning, sequence clustering and OTU assignment, which may significantly affect diversity estimates. Although guidelines exist (Nilsson *et al.*, 2011; Lindahl *et al.*, 2013), and most data treatment pipelines include some sort of quality filtering, a widely accepted consensus on the standardization of fungal metabarcoding data analysis is still lacking. As Lindahl *et al.* (2013) point out, methodological biases can be introduced at different steps during a study, such as sample collection, primer selection, handling of sequencing errors, sequence similarity clustering, and taxonomic assignment of OTUs. This renders data set comparisons problematic, because the choice of methods may significantly affect the

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interpretation of data from environmental metabarcoding. The sources of metabarcoding sequence errors are relatively well characterized. They include PCR misincorporations, platform-related errors in nucleotide detection, and the generation of chimeric sequences (Huse *et al.*, 2007; Dickie, 2010; Edgar *et al.*, 2011; Haas *et al.*, 2011). The incompleteness of public taxonomic databases resulting in unidentifiable sequences is also well known (Kõljalg *et al.*, 2005; Nilsson *et al.*, 2006; Bidartondo *et al.*, 2008). However, it is unclear whether all metabarcoding data sets are affected by these issues to a similar extent. Owing to the lack of comparative studies, we cannot judge the effects of differential data treatment on the estimated diversity and taxonomic composition of fungal communities.

Fungi are key players in ecosystem functioning and services, acting as decomposers, parasites, pathogens or symbionts. Despite their rich biodiversity and high ecological and economical relevance, fungi are still an understudied group of organisms (Pautasso, 2013; Rambold *et al.*, 2013). The magnitude of global fungal diversity is a contentious issue in biodiversity research. What is probably the best-known estimate of fungal diversity, *c.* 1.5 Mio species (Hawksworth, 1991, 2001), was criticized as a result of the small- to large-scale interpolations (May, 1991; debate summarized in Scheffers *et al.*, 2012). May (1991) suggested that there might be far fewer species of fungi. Although the recent estimate of Mora *et al.* (2011) seems to be in line with May, most studies expect high fungal diversity (O'Brien *et al.*, 2005; Bass & Richards, 2011; Blackwell, 2011). This is supported by the recent discovery of higher fungal taxa (Jones *et al.*, 2011) and novel lineages (Amend *et al.*, 2012), the extent of cryptic diversity within morphospecies (Carriconde *et al.*, 2008; Crespo & Lumbsch, 2010), fungal habitat heterogeneity, the shear amount of unexplored habitats, limited primer inclusivity and PCR stochasticity in molecular studies (Schmidt *et al.*, 2013), etc. Estimates of high global fungal diversity are also supported by the increasingly widespread next-generation sequencing metabarcoding studies (Hibbett *et al.*, 2011). Next-generation metabarcoding frequently reveals high taxonomic divergence among geographically close samples (Danielsen *et al.*, 2012; Xu *et al.*, 2012; Schmidt *et al.*, 2013). However, it is still unclear whether the high local diversity scales up to high global diversity. It was shown for tropical insects that high alpha diversity does not necessarily imply high beta diversity (Novotny *et al.*, 2007). In fungi, certain ecological traits, including efficient dispersal, may facilitate occurrence in a wide range of environments and across large geographic distances. Such traits would result in lower global diversity than estimated by local-to-global-scale interpolations. The comparison of fungal molecular diversity based on next-generation sequence data from geographically distant locations may reveal whether widespread fungal lineages and/or habitat generalists exist, and whether high local diversity also implies high global diversity in fungi.

Biogeographic patterns in the diversity and distribution of organisms have long been studied by biologists (Pianka, 1966; Hillebrand, 2004). The decrease of diversity with increasing latitude, and the distance-related decay in community similarity are among the most investigated patterns. Most available data in global biogeography refer to macroorganisms (plants and animals), with relatively few studies on prokaryotes, or fungi. It is important to know whether the biogeographic patterns of microorganisms (including fungi) mirror those of macroorganisms, in order to understand whether natural factors shaping the distribution of biodiversity are universal (Martiny *et al.*, 2006). While a study on the global diversity of foliar endophytes recovered well-known latitudinal biodiversity patterns of macroorganisms (Arnold & Lutzoni, 2007), these patterns were not found in analyses on fungal communities inside buildings (Amend *et al.*, 2010) and ectomycorrhizal fungi (Tedersoo *et al.*, 2012). The increasing availability of next-generation metabarcoding data from environmental samples, and the existence of theoretical frameworks for testing hypotheses on the biogeography of microorganisms (Martiny *et al.*, 2006), improves our chances to detect previously undescribed microbial diversity and to understand biogeographic patterns of fungi.

Here we reanalyzed and compared data from 10 fungal metabarcoding studies. These studies provide deep-sequenced data sets of the internal transcribed spacer (ITS, the selected barcode of fungi; Schoch *et al.*, 2012) generated on 454 platforms. We asked the following questions: what is the effect of different steps in the data pruning process on fungal diversity estimates; how common are widespread fungal species across biomes and habitats; does local fungal diversity scale up to a global level; and are there biogeographic diversity patterns in deep-sequenced fungal communities?

Materials and Methods

Assembly of data sets

This meta-analysis includes sequences from all published papers on 454 (Margulies *et al.*, 2005) environmental metabarcoding of fungal communities (as of September 2012) that fulfilled the following selection criteria: barcode marker is ITS1; sequences were generated on the Roche 454 sequencing platform; and raw sequence data (standard flowgram format (SFF) files) are available. The ITS marker is widely used as barcode to identify fungal species (Schoch *et al.*, 2012), and it is the most common sequence marker in mycology (Begerow *et al.*, 2010). We selected ITS1 because it has been used more often than ITS2 in environmental studies to date. Obtaining SFF files was important because different approaches of pruning and processing raw sequences influence diversity estimates. We found 10 studies matching these criteria: five studies survey fungal communities from soil-associated environments (hereby referred to as 'soil' studies, Buée *et al.*, 2009; Jumpponen *et al.*, 2010; Danielsen *et al.*, 2012; Xu *et al.*, 2012; Yu *et al.*, 2012); four studies target communities in the phyllosphere ('leaf' studies, Jumpponen & Jones, 2009; Cordier *et al.*, 2012; Zimmerman & Vitousek, 2012; Bazzicalupo *et al.*, 2013); and one study focuses on indoor fungi (Amend *et al.*, 2010). We received data sets from the Sequence Read Archive (SRA) of the National Center for Biotechnology Information (NCBI, USA), the European Nucleotide Archive (ENA) of the

Table 1 List of data sources

Study	Fungi	Habitat/host	Geographical region	454 platform	Primers	Internal transcribed spacer (ITS)	Original pruned reads	Tag size (bp)	Raw data sources
Amend *et al*. (2010)	Indoor	Indoor dust	Global	GS FLX Titanium	ITS1-F ITS4	Full	97 577	8	SRA SRS010093
Bazzicalupo *et al.* (2013)	Phyllosphere	*Populus balsamifera* L.	Alaska	GS FLX Titanium	ITS1-F ITS4	Full	141 920	10	ENA ERP001860
Buée *et al.* (2009)	Soil	Temperate forest	France	GS FLX	ITS1-F ITS2	1	166 350	3	Authors
Cordier *et al.* (2012)	Phyllosphere	*Fagus sylvatica* L.	France	GS FLX Titanium	ITS1-F ITS2	1	96 130	5	Authors
Danielsen *et al.* (2012)	Soil/root-associated	*Populus canescens* (Aiton) Sm.	France	GS FLX Titanium	ITS1-F ITS2	1	686 053	10	ENA ERP001442
Jumpponen & Jones (2009)	Phyllosphere	*Quercus* spp.	Kansas	GS FLX	ITS1-F ITS2	1	–	5	Authors
Jumpponen *et al.* (2010)	Ectomycorrhizal	*Quercus* spp.	Kansas	GS FLX	ITS1-F ITS2	1	29 910	5	SRA SRA009804
Xu *et al.* (2012)	Soil/root-associated	*Pisum sativum* L.	Denmark	GS FLX	ITS1-F 58A2R	1	68 811	6	Authors
Yu *et al.* (2012)	Soil/root-associated	*Pisum sativum* L.	Denmark	GS FLX	ITS1-F ITS2	1	60 071	6	Authors
Zimmerman & Vitousek (2012)	Phyllosphere	*Metrosideros polymorpha* Gaudich.	Hawaii	GS FLX Titanium	ITS1-F ITS2	1	665 155	10	SRA SRX153137

ENA, European Nucleotide Archive of EMBL-EBI (European Bioinformatics Institute, part of the European Molecular Biology Laboratory); SRA, Sequence Read Archive of NCBI (National Center for Biotechnology Information).
The data from Jumpponen & Jones (2009) includes an additional tree species which was sequenced, but not discussed in the original publication. Our comparative analysis also included this species.
ITS1-F, Gardes & Bruns (1993); ITS2 and ITS4, White *et al.* (1990); 58A2R, Martin & Rygiewicz (2005).

European Bioinformatics Institute (part of the European Molecular Biology Laboratory, EMBL-EBI) and directly from the authors. An overview of the studies including the fungal habitat, geographical region, primers used and the original number of pruned reads is given in Table 1. Altogether we analyzed > 2.8 million raw reads.

Data pruning

The steps of raw SFF data pruning are summarized in Fig. 1. Data sets were acquired as raw data files in SFF format. Data sets in SRA format downloaded from the NCBI SRA were converted to SFF using sff-dump v.2.1.9 from the NCBI SRA Toolkit and sfffile v.2.7 from the 454 Sequencing System Software Suite v.2.7 (Roche 454 Life Sciences). Pooled samples were extracted into separate SFF files, allowing no mismatches in the identification tag sequence. If such demultiplexed files had > 40 000 reads, they were randomly divided into smaller files in order to use AmpliconNoise v.1.25 (Quince *et al.*, 2011). The binary SFF files were converted into text flowgrams with sffinfo v.2.5.3.

We used AmpliconNoise to remove errors in pyrosequenced amplicons in all studies. According to Quince *et al.* (2011) sources of such errors ('noise') are originating from 454 sequencing itself and from PCR amplification. AmpliconNoise removes sequencing errors and PCR single base errors separately with different subroutines. The first step consists of a flowgram clustering to remove sequencing noise. PyroDist(M) calculates flowgram distances, followed by FCluster, which does a complete linkage hierarchical clustering of the flowgrams. PyroNoise(M) makes use of an expectation-maximization (EM) algorithm to construct denoised sequences. The second step includes a sequence-based clustering to account for nucleotide errors in the PCR process. SeqDist(M) calculates the sequence distances, followed by FCluster for hierarchical clustering. SeqNoise(M) runs an EM algorithm to remove PCR noise by clustering the sequences. AmpliconNoise is incorporated into major pipelines for processing pyrosequencing data (Caporaso *et al.*, 2010; Schloss *et al.*, 2011). Slightly different subroutines from AmpliconNoise were employed for pruning reads from the 454 platform GS FLX and GS FLX Titanium. Thereby distance matrices are generated differently for Titanium reads to fit a slightly different algorithm for the centroid construction to prefer longer sequences for centroid clusters.

We removed chimeric sequences with UCHIME *de novo* (Edgar *et al.*, 2011), a tool from USEARCH v.6.0.203 (Edgar, 2010). Chimeras are artificial sequences that originate from two or more true sequences during PCR amplification. If not identified and removed, chimeric sequences can lead to the false identification of novel taxa and false diversity estimates (Haas *et al.*, 2011). The UCHIME algorithm is faster than other methods, such as Perseus (Quince *et al.*, 2011) or ChimeraSlayer (Haas *et al.*, 2011).

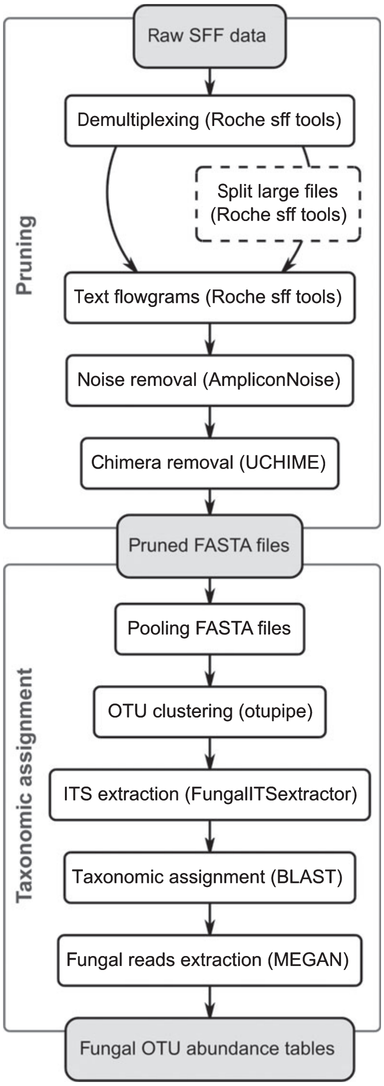

Fig. 1 Analytical pipeline used for data pruning, operational taxonomic unit (OTU) delimitation and taxonomic assignments. Program names are given in brackets. SFF, standard flowgram format; ITS, internal transcribed spacer.

Clustering and taxonomic assignments

The steps for clustering into OTUs and their taxonomic assignment are presented in Fig. 1. To cluster the noise- and chimera-free reads into OTUs, the FASTA files from all studies were pooled into a single file to compare OTUs between different studies. We used the script otupipe v.1.1.9 based on USEARCH v.6.0.203 (Edgar, 2010) to group the reads into OTUs. The script makes use of the fast UCLUST algorithm to divide a set of sequences into clusters. Here, we chose 97% as the similarity threshold for clustering as the most relevant one in the taxonomic context. This cut-off has often been used in fungal community analyses (Nilsson *et al.*, 2008; Tedersoo *et al.*, 2010). However, no generally preferable standard for meaningful OTU delimitations is accepted (O'Brien *et al.*, 2005; Hibbett *et al.*, 2011). OTUs generated with different clustering algorithms are not directly comparable.

The ITS1 forward reads were filtered with the FungalITSExtractor v.2 from Nilsson *et al.* (2010). This also excludes sequences that are not ITS1. ITS1 is a highly variable region, flanked by the conserved gene for the small subunit (SSU) and the conserved 5.8S gene. Removing the conserved DNA fragments is important, as they might otherwise result in incorrect assignments during sequence similarity searches (Nilsson *et al.*, 2009).

To identify the extracted ITS1 OTUs, we performed a BLAST v.2 search (Altschul *et al.*, 1990) against the entire GenBank (Benson *et al.*, 2013) BLAST database downloaded from NCBI on 18 October 2012. We used MEGAN v.4.69.4 (Huson *et al.*, 2011) with the latest version of the NCBI taxonomy, updated on 27 November 2012, to classify the sequences according to their taxonomic context. MEGAN estimates and explores the taxonomic content of a data set by parsing BLAST output files and assigning each sequence to the lowest common ancestor (LCA) taxon in the NCBI taxonomy. Only alignments with a minimum bit-score of 170, matches within 5% of the highest score and a minimum of one read were considered for the taxonomic assignment. Sequences classified as fungi were retained for further analysis.

Rarefaction curves and shared OTUs

Rarefaction analysis was used to assess the taxonomic richness of the samples. Richness estimates are strongly dependent on sampling intensity, as the probability of detecting more and rare species increases with sampling effort (Magurran, 2004). The rarefaction technique allows meaningful standardization to avoid incompatibility of collected data because of different samples sizes (Gotelli & Colwell, 2011). We calculated rarefactions in mothur v.1.24.1 (Schloss *et al.*, 2009) with the command rarefaction.single with a frequency of 1000. This subroutine randomly resamples the pool of sequences multiple times without replacement and calculates the statistically expected number of OTUs found in each resampled subset. The rarefaction curves were plotted in R v.2.15.3 (R Core Team, 2013).

We counted OTUs shared among all studies, among phyllosphere studies, and among soil studies. OTUs shared in at least three phyllosphere studies and at least three soil studies were taxonomically identified with BLAST, followed by MEGAN. We visualized the accumulation of novel OTUs with an OTU accumulation curve using R with the specaccum command of the vegan package v2.0-7 (Oksanen *et al.*, 2013). Error margins were estimated with 100 random permutations. To test the robustness of our findings concerning OTU sharing, we analyzed the data also at more relaxed OTU similarity thresholds (95 and 90% sequence similarity).

Biogeographic analyses

For biogeographic analyses, we randomly subsampled 15 900 reads (the lowest number of reads available from a single study) from each of the nine plant-associated read sets to account for the differences in read numbers. We omitted the study on fungal communities in indoor dust (Amend *et al.*, 2010) from these analyses. All calculations were done after the removal of singletons (OTUs represented by only one sequence) from the OTU abundance matrices. We tested the relationship between OTU richness and latitude. We fitted a generalized linear model (GLM) of the gamma family with a logarithmic link on the nine compared fungal richness data sets in R. We accounted for study-specific differences of the communities by including the number of distinct fungal groups/habitats identified by the individual studies as an independent explanatory variable. For example, Bazzicalupo *et al.* (2013) report three distinct, host genotype-related fungal community groups, while Zimmerman & Vitousek (2012) report 13 strongly different communities that correspond to variations in precipitation, elevation and nutrient availability at 13 sampling sites. We also included the type of the fungal community (associated with leaves or soil) as an independent explanatory variable in the GLM.

The compositional similarity of the plant-associated fungal communities was evaluated with the Jaccard index. Presences and absences of nonsingleton fungal OTUs were used to calculate the Jaccard similarity in R with the vegdist command of the vegan package. We calculated the global (Haversine) distances among the nine study sites in R using the package geosphere v1.2-28 (Hijmans *et al.*, 2012). As some studies did not share any OTUs, the distance-related decay in community similarity could not be estimated with a linear regression of log-transformed similarity values against geographic distance (Nekola & White, 1999). We dealt with zero-similarity studies by implementing a GLM on the community similarity data against geographic distance. The model had binomial observation error and a log-link function (Millar *et al.*, 2011). As the site pairs are not independent, the error estimates from the model are not valid. We obtained valid standard errors of the estimated parameters by a leave-one-out jacknife approach using a script provided by Millar *et al.* (2011).

Results

Data pruning

Pruning resulted in differential loss of reads in the individual studies (Table 2). Demultiplexing without allowing mismatches in the tag sequence resulted in 1–3% fewer raw reads, except for the Cordier *et al.* (2012) and Amend *et al.* (2010) studies where read numbers were reduced by 10 and 49%, respectively. AmpliconNoise regarded between 11 and 53% of the raw reads as PCR or sequencing noise (Table 2). UCHIME found an average of 1% (0.1–3.9%) chimeric sequences in the studies. Pruning of the raw data sets, including noise and chimera removal, eliminated about one-third of the raw reads on average. This is roughly equal to the amount of pruned reads in the original studies (Table 1), although the pruning used here is not the same as in all original studies. The 62% sequence loss for the Amend *et al.* (2010) study is mainly the result of the stringent demultiplexing. The data set from Cordier *et al.* (2012) had 55% sequencing noise, primarily removed during demultiplexing and noise removal. Yu *et al.* (2012) also had 55% sequencing noise, but mainly from noise removal alone.

Clustering and taxonomic assignments

Clustering the pruned and combined data sets from all studies with otupipe resulted in 30 929 OTUs at a similarity threshold of 97%. Of these, 27 056 OTUs belonged to the ITS1 region. Taxonomic assignment in BLAST and MEGAN yielded 10 495 fungal OTUs. Overall, about one-third of the OTUs were singletons. The OTU numbers for each study are given in Table 3, the corresponding read numbers are presented in Table 2. Both the numbers of OTUs and reads differ strongly among studies. Clustering at 97% sequence similarity yielded 1000–2000 OTUs in most studies, but more than twice as many in some (Table 3). Corresponding read numbers range from 23 975 up to 666 746 in these studies.

The extraction of ITS1 with the FungalITSExtractor did not substantially reduce the number of OTUs or reads. About one-fifth of OTUs and reads were discarded in the studies of Amend *et al.* (2010), Bazzicalupo *et al.* (2013) and Danielsen *et al.* (2012), whereas the proportion was about one-tenth in the other studies.

Less than half of all ITS1 OTUs were identified as fungi during the taxonomic identification with BLAST and MEGAN. Generally *c.* 58% of the OTUs could not be assigned taxonomically and *c.* 3% belonged to other eukaryotes and prokaryotes. One-quarter of all OTUs were identified as Ascomycota, mainly Saccharomyceta. The proportion of Basidiomycota was *c.* 8%,

Table 2 Decline of read numbers during data pruning, after 97% similarity clustering, internal transcribed spacer 1 (ITS1) extraction, and the selection of fungal operational taxonomic units (OTUs)

	Amend *et al.* (2010)	Bazzicalupo *et al.* (2013)	Buée *et al.* (2009)	Cordier *et al.* (2012)	Danielsen *et al.* (2012)	Jumpponen & Jones (2009)	Jumpponen *et al.* (2010)	Xu *et al.* (2012)	Yu *et al.* (2012)	Zimmerman & Vitousek (2012)
Raw	288 360	204 052	185 806	141 270	–	29 643	33 656	78 244	282 115	778 476
Demultiplexed	146 683	199 526	180 213	126 451	811 900	29 155	32 836	76 842	276 438	771 223
AmpliconNoise	111 745	165 292	145 330	64 075	531 026	24 736	24 000	61 663	128 655	688 875
UCHIME	110 351	158 769	144 857	63 793	529 687	24 696	23 975	61 527	126 520	666 784
97% clustering	110 295	158 758	144 850	63 793	529 672	24 696	23 975	61 527	126 506	666 746
IST1 extraction	103 448	141 074	144 511	63 625	524 919	24 672	23 949	59 439	126 047	614 616
Fungal reads	19 159	69 597	128 305	53 953	448 464	15 926	18 382	40 644	112 950	284 439

The raw data from Danielsen *et al.* (2012) were provided as demultiplexed raw files.

Table 3 Operational taxonomic unit (OTU) numbers after 97% sequence similarity clustering, internal transcribed spacer 1 (ITS1) extraction, and discarding nonfungal OTUs

	Amend *et al.* (2010)	Bazzicalupo *et al.* (2013)	Buée *et al.* (2009)	Cordier *et al.* (2012)	Danielsen *et al.* (2012)	Jumpponen & Jones (2009)	Jumpponen *et al.* (2010)	Xu *et al.* (2012)	Yu *et al.* (2012)	Zimmerman & Vitousek (2012)
97% OTUs	4378	1881	1743	1466	7799	1322	1037	1626	1986	8964
ITS1 OTUs	3494	1574	1658	1410	6297	1309	1026	1477	1937	8701
Fungal OTUs	194	563	945	842	2467	757	631	657	1152	3602

Table 4 Number of shared operational taxonomic units (OTUs) at 97% similarity clustering

Shared OTUs in number of studies	1	2	3	4	5	6
All studies (10)	9410	900	149	28	7	1
Phyllosphere studies (4)	5224	232	24	1		
Soil studies (5)	4735	467	57	3	–	

and Glomeromycota and early diverging fungal lineages were represented by *c.* 1%. Altogether, filtering for ITS1 and fungi reduced the OTU numbers in the data sets by 39–96%. Amend *et al.* (2010) had one of the highest numbers of OTUs before filtering, but the lowest after filtering.

Community similarity and biogeography

Few OTUs (at 97% similarity) are shared among studies (Table 4). About 90% of the fungal OTUs are present only in a single study. This resulted in a steep OTU accumulation curve (Fig. 2a). The plant-associated fungal communities were highly dissimilar (Fig. 2b), sharing only *c.* 1.6% of the species on average. The highest proportion of unique OTUs was found in Zimmerman & Vitousek (2012) (94%), Buée *et al.* (2009) (83%) and Danielsen *et al.* (2012) (80%). There is only one OTU shared among six studies, and seven OTUs are shared among five studies. These could only be assigned to higher-level taxa using the ITS1 fragment: fungi, two Ascomycota, Erythrobasidiales, two Pleosporaceae and two leotiomyceta. Even when comparing only the soil studies, or only the phyllosphere studies, few OTUs are found in more than two data sets (Table 4). Only one OTU is present in all four phyllosphere studies and 24 OTUs in three out of four studies. No OTUs are shared among all five soil studies, three OTUs are shared in four soil studies and 57 OTUs in three of the studies. We selected all OTUs shared among at least three phyllosphere studies and at least three soil studies, and presented them in a phylogenetic tree derived from taxonomic assignment in MEGAN (Fig. 3). Only 21 of the 85 shared OTUs could be assigned to species or genus level; the rest were assigned to higher-level taxa. A brief characterization of their lifestyle is provided in Table 5. Most of these shared OTUs are widespread according to Kirk *et al.* (2008) and represent mainly saprotrophic and potentially pathogenic taxa. Using less stringent clustering thresholds (95 and 90%) resulted in only slightly increased numbers of shared OTUs (Supporting Information, Table S1).

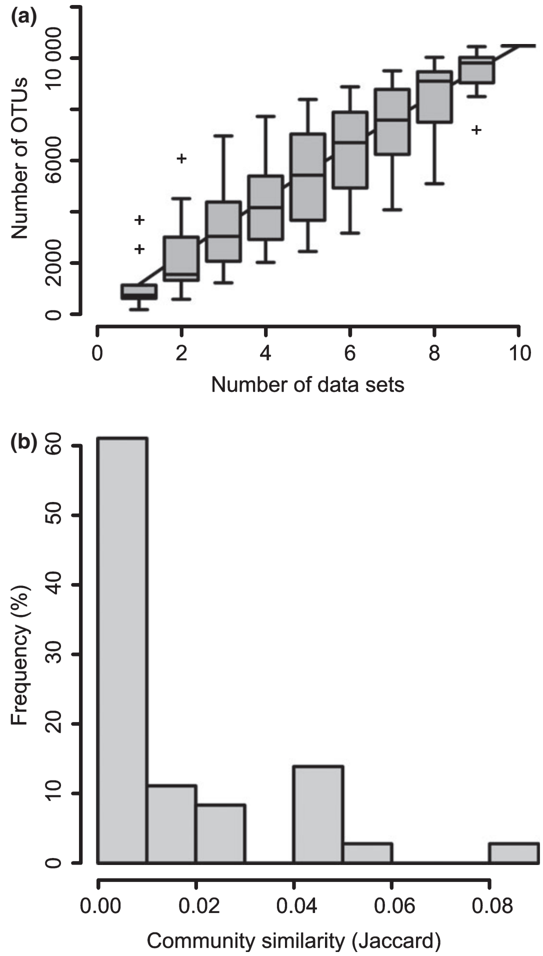

Fig. 2 Fungal communities are highly dissimilar among compared studies. This is indicated by (a) the accumulation curve of fungal operational taxonomic units (OTUs) (boxplots mark standard deviations, + signs represent outliers); and (b) the distribution of Jaccard similarities among study pairs.

The taxonomic composition in soil and phyllosphere studies is given in Fig. 4. We show the distribution of relatively common OTUs among fungal classes (classes represented by > 1% of the total OTUs). Phyllosphere fungi occur in 10 currently accepted classes, soil fungi in 13. We found more OTUs assigned to Ascomycota than to Basidiomycota and other basal fungal lineages in the present data. The most OTU-rich classes in the phyllosphere are Dothideomycetes and Sordariomycetes, accounting for 65% of all phyllosphere OTUs. The most

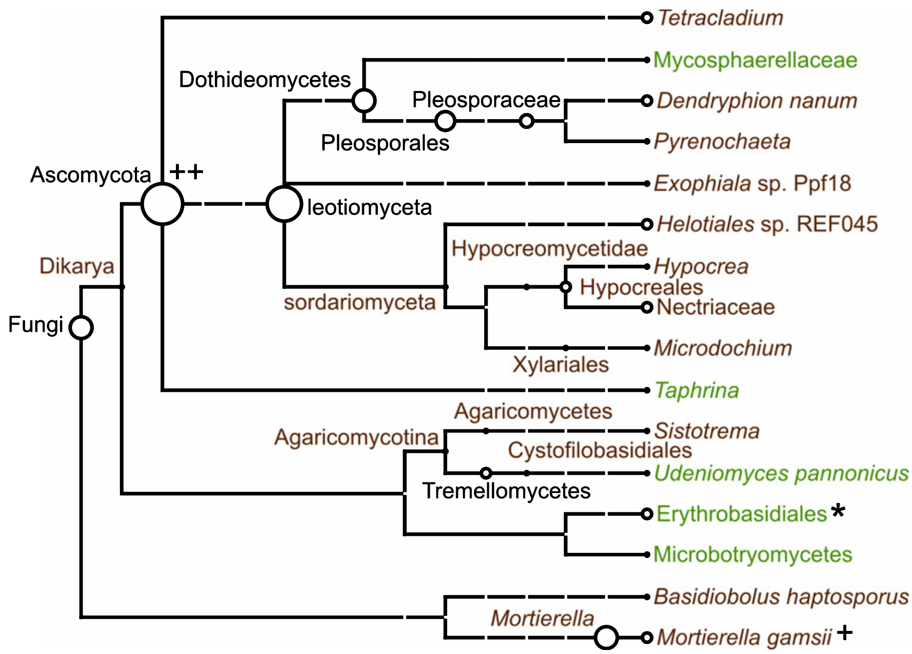

Fig. 3 Taxonomic assignment of the most frequently shared operational taxonomic units (OTUs). OTUs encountered in more than three phyllosphere or soil studies are presented. Circles represent taxonomic ranks with assigned reads, and their size is scaled according to the number of OTUs assigned to that unit. Green, OTUs from phyllosphere studies; brown, OTUs from soil studies; black, OTUs from both phyllosphere and soil studies. *, OTUs present in all four phyllosphere studies; +, OTUs present in four out of five soil studies.

OTU-rich classes in the soil are Agaricomycetes and Sordariomycetes, accounting for 52% of the soil OTUs. The total number of fungal OTUs assignable to classes in the phyllosphere is 3857, in soil 3440.

Rarefactions curves for all studies are shown in Fig. 5. The newer studies by Danielsen *et al.* (2012) and Zimmerman & Vitousek (2012) show longer curves than the rarefactions of the other studies, indicating a deeper sampling effort. Some of the rarefaction curves seemingly approach saturation, but without reaching a plateau, for example, Bazzicalupo *et al.* (2013) or Amend *et al.* (2010), while others end in steep slopes (Jumpponen & Jones, 2009; Jumpponen *et al.*, 2010).

The OTU richness among the nine compared fungal communities was significantly explained by latitude (log-scale intercept $a = 6.94$, log-scale slope of latitude $b = -0.02$, $P = 0.03$; Fig. 6a). The complexity of the individual studies (e.g. whether data were collected along environmental gradients) and the type of community (phyllosphere or soil-associated fungi) had no significant explanatory value. The similarity of the communities decreased slightly with increasing geographic distance according to the GLM (Fig. 6b), with an intercept at $a = -3.855$ (standard error (SE) = 0.5, values expressed as logarithmized similarity) and a slope of $b = 0.00005$ (SE = 0.00009, values expressed as logarithmized similarity decrease per 1 km).

Discussion

High-throughput metabarcoding revolutionizes our understanding of fungal diversity and community composition. However, different researchers have used different data pruning approaches and clustering thresholds, thus precluding direct comparison of OTUs and taxonomic composition among studies. By applying the same pruning methods to 10 published data sets, we showed highly variable noise levels (between 11 and 53%, Table 2). Interestingly, the smallest portion of sequencing and PCR noise was found in the largest and most OTU-rich study (Zimmerman & Vitousek, 2012). This is in contrast with Bakker *et al.* (2012), who showed that more diverse communities had higher AmpliconNoise implied error rates than low diversity communities. However, the effects of denoising on output data quality and quantity are still not fully understood. Gaspar & Thomas (2012) found that the denoising process can make inconsistent changes to sequence reads, resulting in unexpected and significant effects on the output of the pipelines. It is plausible that there is considerable stochasticity in the noise levels produced in different 454 sequencing runs, the source of which is not well understood. We found the largest discrepancies in denoising results between the original studies and our analysis in those studies that did not employ an algorithmic denoising tool, but used simple quality filtering (Cordier *et al.*, 2012; Danielsen *et al.*, 2012; Yu *et al.*, 2012). Chimera checking indicated *c.* 1% of chimeras in the data sets (Table 2). We thus cannot confirm the observation that diversity estimates are mainly inflated by chimeric sequences formed especially in more diverse samples (Fonseca *et al.*, 2012). In the 10 compared studies, the removal of sequencing and PCR errors (denoising) had a greater influence on the data, and thus on downstream analyses, than chimera removal. However, the low proportion of chimeras found in the data sets could also result from the inability of tools to detect them.

Table 5 Taxa assignable to species or genera, shared among at least three phyllosphere or three soil studies

Taxon name	Division	Order	Lifestyle	GenBank taxon id	Shared in	References[a]
Basidiobolus haptosporus Drechsler	Entomophthoromycota	Basidiobolales	Saprotrophic, biotrophic on mites	90248	Soil	1, 2, 3, 15
Dendryphion nanum (Nees) S. Hughes	Ascomycota	Pleosporales	Endophytic, potential plant pathogen	256645	Soil	6, 7
Exophiala sp.	Ascomycota	Chaetothyriales	Saprotrophic, endophytic; opportunistic human pathogens	703958	Soil	4, 14
Helotiales sp.	Ascomycota	Helotiales	Mostly saprotrophic, many species necrotrophic, hemibiotrophic or biotrophic plant pathogens; few lichenized or lichenicolous	1160011	Soil	5, 14
Hypocrea sp.	Ascomycota	Hypocreales	Saprotrophic on decaying woody substrata, sometimes on old fruit bodies of other fungi, in soil, litter; in asexual state (Trichoderma) eventually mycoparasitic	29859	Soil	5, 11, 12, 14
Microdochium sp.	Ascomycota	Xylariales	Saprotrophic or parasitic; on litter, roots, grasses (pathogenic on cereals), and other fungi	67608	Soil	5, 12
Mortierella gamsii Milko	Incertae sedis ('Zygomycota')	Mortierellales	Saprotrophic in soil	64522	Soil	3, 16
Mortierella sp.	Incertae sedis ('Zygomycota')	Mortierellales	Saprotrophic in soil	4855	Soil	3, 14
Pyrenochaeta sp.	Ascomycota	Pleosporales	Many pathogens of grasses; opportunistic human pathogens	285810	Soil	4, 5
Sistotrema sp.	Basidiomycota	Corticiales	Saprotrophic on wood and other plant material; ectomycorrhizal	139131	Soil	1, 8
Taphrina sp.	Ascomycota	Taphrinales	Biotrophic on plants (mostly Fagaceae and Rosaceae)	5010	Phyllosphere	3, 5, 14
Tetracladium sp.	Ascomycota	Helotiales	Aquatic, endophytic	164535	Soil	10, 13
Udeniomyces pannonicus Niwata, Tornai-Leh., Deák & Nakase	Basidiomycota	Cystofilobasidiales	Saprotrophic, isolated from leaves	184025	Phyllosphere	9

[a]References: 1, Ainsworth *et al.* (1973); 2, Campbell *et al.* (2013); 3, Cannon & Kirk (2007); 4, Desnos-Ollivier *et al.* (2006); 5, Kirk *et al.* (2008); 6, Lenc *et al.* (2011); 7, Lenc *et al.* (2012); 8, Marino *et al.* (2008); 9, Niwata *et al.* (2002); 10, Roldán *et al.* (1989); 11, Rossman *et al.* (1999); 12, Seifert *et al.* (2011); 13, Selosse *et al.* (2008); 14, Webster & Weber (2007); 15, Werner *et al.* (2012); 16, Zycha & Siepmann (1969).

The use of the different 454 chemistries (FLX and Titanium) translates into different read lengths, which may introduce some bias during taxonomic assignment. However, we assumed that the assignments are not affected considerably, because the FLX (the older chemistry) already provides reads that cover the complete ITS1 region in most cases, and only ITS1 was used for taxonomic assignments. Taxonomic identification with BLAST revealed a large portion of unidentifiable OTUs. More than half of all ITS1 OTUs clustered at 97% similarity were not assignable to any taxa in NCBI. Sequences not matching GenBank records might represent novel species, or described, but unsequenced, species (Nagy *et al.*, 2011; Hibbett & Taylor, 2013). While some researchers prefer to use well-documented and custom-curated databases such as UNITE (Abarenkov *et al.*, 2010) or RefSeq (Pruitt *et al.*, 2012), we opted for GenBank (Benson *et al.*, 2013), because MEGAN assignments are based on GenBank sequence headers. Problems arising from poor-quality entries may be circumvented by using the lowest common ancestor-based taxonomic assignment of BLAST hits, for example, the algorithm in MEGAN (Huson *et al.*, 2011). Besides OTUs that were not assignable at all, only 3% of the OTUs were assigned to nonfungal taxa. A relatively high proportion of nonfungal sequences were found in those studies targeting the full ITS region and using the nonfungal-specific primer ITS4 (Amend *et al.*, 2010; Bazzicalupo *et al.*, 2013). Recently, high-coverage fungal-specific ITS primers have been developed (Ihrmark *et al.*, 2012; Toju *et al.*, 2012), which may contribute to a more targeted and exhaustive sampling of environmental fungal diversity, and avoidance of nonfungal DNA.

The low number of shared OTUs among studies suggests high global fungal diversity. Most sample pairs were highly dissimilar, having < 0.5% of fungal OTUs in common (Fig. 2a,b).

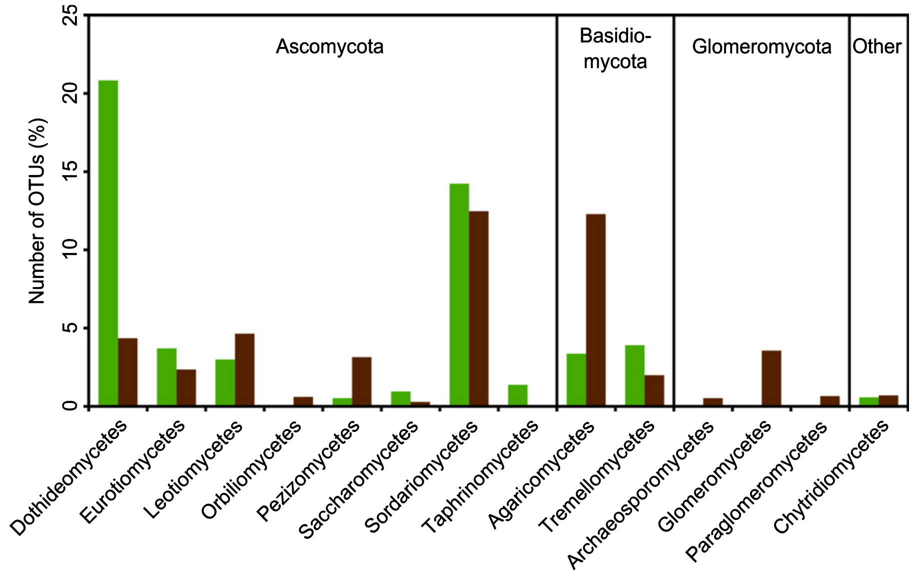

Fig. 4 Fungal classes identified in the soil and phyllosphere studies compared. Bars represent the relative abundance of operational taxonomic units (OTUs) assigned to a fungal class (phyllosphere, green; soil, brown).

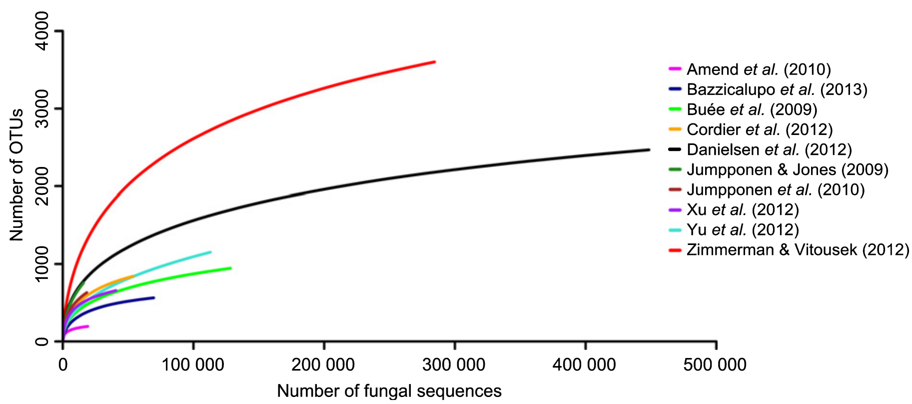

Fig. 5 Rarefaction curves of the compared studies. The curves were calculated using pruned, internal transcribed spacer 1 (ITS1) fungal sequences. Operational taxonomic units (OTUs) were assigned at 97% sequence similarity.

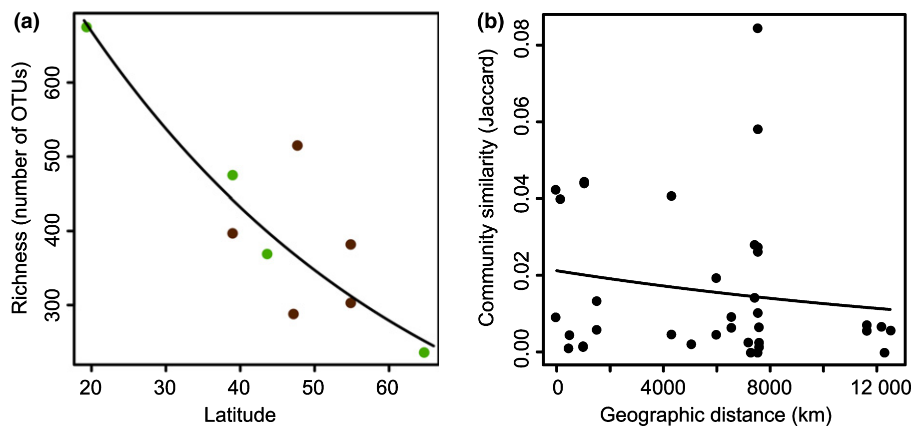

Fig. 6 Biogeographic patterns in plant-associated fungal communities. (a) Latitudinal gradient in the operational taxonomic unit (OTU)-based richness of compared communities (phyllosphere, green; soil, brown). (b) Distance decay in the similarity of fungal communities.

Even the same habitat type (phyllosphere or soil) did not show many shared OTUs. According to Bass & Richards (2011), it is still largely unknown how much sampling (sequencing) effort is required to saturate species accumulation curves even if samples are geographically close. In line with this note, sampling was unsaturated in each of the compared studies (Fig. 5). We cannot exclude the possibility that further increasing the sampling depth may identify more shared OTUs, but we emphasize that the compared studies represent the deepest sample sets that characterize fungal diversity to date. Only few, relatively common fungal lineages (Fig. 3) could be assigned to species or genus level. This suggests that even the more widespread fungal lineages are poorly known, or that they are missing from sequence databases. Most shared, identifiable fungal OTUs were assigned to taxa that include known saprotrophic or pathogenic lineages. Considering that we obtained 10 495 fungal OTUs from only 10 studies by employing a rather conservative data cleanup procedure, and found almost no shared OTUs among these studies, we conclude that species-level diversity in fungi is probably at the higher end of most current diversity estimates. This finding is corroborated by an additional analysis using more liberal OTU assignment thresholds (95 and 90% similarity), which yielded only a few more shared OTUs (Table S1).

Operational taxonomic units of various fungal groups were not evenly distributed among fungal classes in this meta-analysis, or in soil vs phyllosphere studies (Fig. 4). Overall, the most OTU-rich fungal classes were Sordariomycetes (27% of all fungal OTUs assignable to classes from the 10 original studies), Dothideomycetes (25%), Agaricomycetes (16%), Leotiomycetes (8%), Eurotiomycetes (6%), and Tremellomycetes (6%). Sordariomycetes is the only group showing high diversity in both phyllosphere and soil environments, indicating the high ecological plasticity of the group. Indeed, described members of the Sordariomycetes occur in all ecosystems, and function as plant and animal pathogens, endophytes of plants, and saprobes involved in decomposition and nutrient cycling (Zhang *et al.*, 2006). Agaricomycetes have been reported to be diverse in soil ecosystems (Buée *et al.*, 2009), which is confirmed here. We also show that 3% of the phyllosphere OTUs belong to Agaricomycetes. The phyllosphere diversity is dominated by Dothideomycetes. Only 4% of the soil fungi belong to this class. Taphrinomycetes are restricted to the phyllosphere, while Leotiomycetes and Glomeromycota are only found in the soil. Most OTUs shared among at least three studies could only be assigned to higher taxonomic ranks (Fig. 3). The inability to safely assign the most widespread lineages indicates large gaps in the knowledge of global fungal diversity. This gap hinders ecological interpretations, and emphasizes the urgent need to develop guidelines for next-generation sequencing taxonomy (Hibbett & Taylor, 2013).

Our results indicate that the compared plant- and soil-associated fungal communities may have biogeographic patterns similar to those of macroorganisms. The richness of fungal communities strongly decreases with increasing latitude. This is in accordance with the long-recognized latitudinal patterns of macroorganisms (Pianka, 1966; Hillebrand, 2004). It also confirms the findings of Arnold & Lutzoni (2007) on diversity patterns based on cultured foliar endophytic fungi. However, there are known exceptions from the latitudinal diversity gradient, including studies on indoor fungi (Amend *et al.*, 2010), and ectomycorrhizal fungi (Tedersoo *et al.*, 2012). The contradiction may be partly explained by the artificial environments of indoor fungi, and the restricted geographic sampling of the soil studies included here. All soil samples originated from similar, intermediate latitudes and thus do not show the previously reported unimodal diversity distribution with the highest diversity at intermediate distances from the equator (Tedersoo *et al.*, 2012). However, the GLM we fitted on the diversity of samples suggested that the type of study system (leaf or soil-associated fungi) had no effect on the distribution of diversity, the only significant explanatory variable being the latitude of the samples. However, this conclusion is based on a sparse data matrix and further effort is necessary to evaluate these relationships better.

The nine plant- and soil-associated fungal communities showed great differences, and their similarity decreased further with increasing geographic distance. Distance decay in the similarity of communities of macroorganisms is frequently reported (Millar *et al.*, 2011). However, much less is known about the effect of distance on the similarities of microbial communities. Green *et al.* (2004) report decay in the similarity of fungal communities at continental scales, while Schmidt *et al.* (2013) report distance decay at only a few meters. Similarly to macroorganisms, the distance-related decrease in the similarity of fungal communities suggests that historical factors, such as dispersal limitations and past environmental conditions, may have influenced the composition of these communities (Martiny *et al.*, 2006). However, we must consider that the communities compared here were highly dissimilar (Fig. 2b), so that the detected similarity decrease could not be very strong. Our results call for more detailed metabarcoding-based analyses of global trends in fungal diversity, with patterns related to habitats and ecological functions specifically considered.

Conclusions

Our study is the first comparison of published and compatible metabarcoding data sets of fungal communities mostly from phyllosphere and soil environments, with clear strengths (deeply sequenced and comparable communities) and weaknesses (scarce spatial resolution, incomparable environmental data). The compared studies contain different levels of noise originating from several sources. This indicates strong methodological stochasticity of environmental metabarcoding. OTU sharing among the deep-sequenced data sets was very limited, suggesting that there are no globally distributed taxa in soil and phyllosphere. Instead, each study appears to recover distinct communities defined by local environments. The few OTUs that were shared among several studies could generally be assigned only to higher taxa, but not to species or genera. This suggests that the high local diversity of fungal communities probably scales up globally, and this diversity is poorly characterized taxonomically and functionally. Our comparison also points toward global trends in the composition and similarity of plant-associated fungal communities. Community diversity is higher towards the equator, and community divergence further increases with geographic distance, with most taxa being locally distributed. These trends are surprisingly similar to the trends long recognized for most macroorganisms. We suggest that global trends in fungal biogeography should be tested with designated large-scale studies. These studies should incorporate well-established hypotheses about the global patterns of macroorganismic diversity (Willig *et al.*, 2003; Martiny *et al.*, 2006). We are convinced that microorganisms provide excellent, but so far underexplored, opportunities to test hypotheses about the distribution patterns of global biodiversity.

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Supporting Information

Additional supporting information may be found in the online version of this article.

Table S1 Number of shared operational taxonomic units (OTUs) at different thresholds of sequence similarity

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